

SmartHouse Documentation

DEMKit 2 edition

The smarthouse demo runs on the DEMKit software, using some specific configuration files on a Raspberry Pi 2. These are included in the Git repository of DEMKit. Additionally, OpenHAB is used as an user interface, while Grafana is used to display graphs. This document briefly describes how to start the Smarthouse demonstration.

PREPARING THE HOUSE

- Make sure all cables are connected:
- Raspberry PI: Ethernet (to router), Hardware interface, USB power cable connected to the PSU, and SD-card inserted
- Power supply: Power cord, make sure it is turned on and USB cable inserted
- Adapter to the model and ethernet connection
- Disable WIFI on other equipment. The Raspberry PI needs to boot first such that it obtains the default IP of 192.168.2.100.

CONNECT THE POWER PLUG

Power the house. Lights should blink on all the hardware. After some time, the Raspberry PI should be booting, which is indicated by a light show in the house where each light blinks a couple of times, and then fades in. After all lights are controlled, the system waits till all services are started. When this is done, all lights will turn on the house and gradually fade out.

When the house is dark again, the system is ready to be used and should have the default IP of 192.168.2.100. It is now safe to turn on the WIFI on other devices and connect to:

SmartHouseNet (no password required)

OPENING OPENHAB

To show the control interface, e.g. on a tablet, open a browser and go to: <http://192.168.2.100:8080>. Choose for the Basic UI.

OPENING GRAFANA

To show graphs, browse with a browser to <http://192.168.2.100:3000> and login (user=admin, pass=admin). Click on the starred dashboard (Utwente / CAES Smart House). It should refresh every 1s.

RUNNING THE DEMO

By default, the simulation is paused. To start, hit the PLAY-button in OpenHAB. The fridge nicely serves as a heartbeat to indicate that the system is still running.

TROUBLESHOOTING

Usually, there should be no problem to start the demo. If there are errors, make sure that:

1. The whole house is booted before connecting other equipment to the WIFI!
2. All cables are connected to the raspberry pi: Ethernet, hardware interface, SD-card and ofcourse power
3. The Router is also powered
4. The house boot animation lights up.
5. Make sure all devices are connected to SmartHouseNet WIFI.

If it still does not boot, then use a recovery SD card!

Furthermore, the house also has indication leds during the boot. After the individual light test, the 2 TVs and charging pole will blink, indicating that services are starting. If a service is started, one of these indicators becomes solid, with the following mapping:

Charging pole: OpenHAB service

TV Living (white): InfluxDB

TV Backroom (green): Grafana

PASSWORDS

Raspberry Pi:

user=openhabian

password=openhabian

Grafana:

user=admin

password=admin

REINSTALL

In case of a reinstallation, it is wise to take the OpenHABian distribution and clone DEMKit. Install the required python packages, InfluxDB and Grafana. Copy the OpenHAB config files (found in the Smarthouse DEMKit model) into /etc/openhab2 folder. Furthermore, make sure that DEMKit is autostarted upon boot. This can be done by using:

\$ sudo crontab -e

An example shell script is available. Furthermore, the Grafana dashboard XML can be imported.